Research Summary

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Catalysts work under potential, solvent, adsorbates and electric fields that reshape their structure and electrostatics, yet most models treat them as static and unpolarized. My research asks **how these operating conditions influence their structure and reactivity**, and builds the simulation, machine-learning and AI tools to answer these questions.

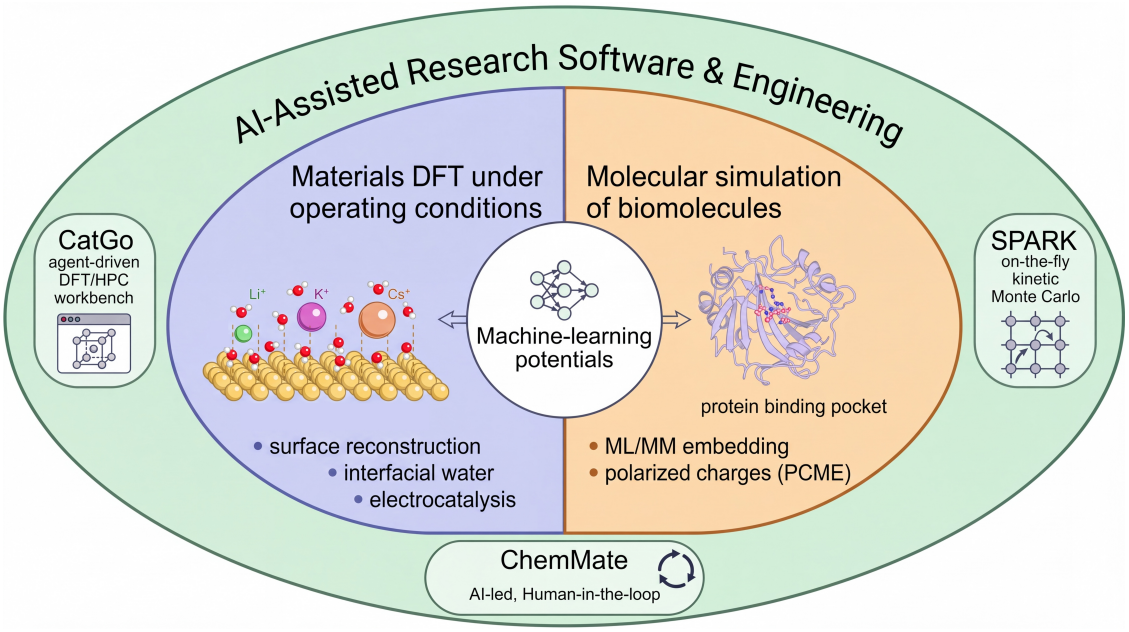


Figure 1. Two research interests linked by machine-learning interatomic potentials, built on AI-assisted research infrastructure.

(a) Surface reconstruction and interfacial structure — *what does a working interface look like?*
I started my research with the interfacial solvent: ab initio MD showed that surface defects anchor interfacial water and raise its cavitation free energy, gating hydrophobic transport to the surface (*Chem. Sci.* **2026**, *17*, 15735–15744). The same water structure explained a puzzle in cation effects: Au surface geometry templates interfacial water enough to reverse the cation trend in CO_2 reduction (*Angew. Chem. Int. Ed.*, under review). Turning from the solvent to the catalyst itself, I used constant-potential ensembles to show that the size effect of IrO_x clusters in acidic OER is set by the populations of local Ir–oxo environments (*Adv. Funct. Mater.*, under review).

At UCLA, I am taking the next step toward surfaces that restructure under reaction, using site-resolved KMC simulation to identify which kinetically trapped, restructured Cu sites dominate the C–C coupling rate.

(b) Catalytic mechanisms and design — *from active site to overall activity*
At Fudan University, I tested a Zn–O–Fe photocatalyst and used DFT to reveal that the medium Fe spin state selects $\bullet CH_2OH$ over $\bullet OCH_3$, steering CH_4 oxidation to ethanol (*Angew. Chem. Int. Ed.* **2025**, *64*, e202510241). I then turned mechanism into design, using machine-learning methods—a stability-first screen and an interpretable SISSO descriptor—to rank doped MoS_2 for nitrogen reduction over HER (to be submitted).

(c) AI-assisted scientific software — *making these campaigns possible in this AI era*
In previous projects, ensembles and screening involved thousands of calculations across computing clusters. To speed up these processes, I co-built the infrastructure: CatGo lets AI coding agents drive an interactive structure/DFT/HPC workbench with every action reviewable (*Digit. Discov.* **2026**, *5*, 3825–3849); ChemMate builds a human-in-the-loop infrastructure for AI-assisted research across the full research lifecycle, in which AI carries out the steps and researchers make the key decisions (ChemRxiv 2026).

(d) Molecular modeling: electrostatics in enzyme design — *exploring beyond heterogeneous catalysis*
The environment matters just as much in biological systems: environment-induced polarization governs protein–ligand binding. To describe the binding energies of protein–ligand complexes more physically, I developed polarization-corrected mechanical embedding (PCME) and incorporated it into PBSA calculations, enabling fast binding-energy calculations (to be submitted).

Next — *from structure to autonomous discovery*
As AI advances, routine, template-like computational chemistry is losing its value. I would like to explore two directions during my PhD. First, catalysts under operating conditions: the interfacial water environment, and the real operando structures and reaction mechanisms of reconstructed catalysts.

Second, AI-driven labs for autonomous discovery. AI for science needs to connect with real experiments. I want to build labs combining multimodal models and robotics with humans in the loop.